**Done by : Faris Tahaineh**

**1- Which of the following segregations could never result in a live birth ?**

a. Tertiary monosomy

b. Interchange monosomy

c. Interchange trisomy

d. Tertiary trisomy

e. All have been reported in live birth babies

**2-In predicting segregant outcomes of reciprocal translocations , we assume that alternate segregationis frequent in gametes. Choose the correct statement :**

a. Adjacent 1 segregation has the highest frequency of all

b. Alternate segregation is reported to be more frequent than all other segregations combined

c. Alternate segregation is most frequent but less than frequency of all segregations combined

d. Adjacent 2 frequency has the lowest frequency of all

e. All segregations have equal chance of occurring

**3- Choose the correct statement regarding assessing phenotypic effect of chromosomal imbalance :**

a. HAL imbalance of less than 1% is always benign

b. HAL imbalance of greater than 2% is always lethal

c. All choices are correct

d. Haploid autosomal length HAL is the gold standard for assessment

e. HAL is a rough tool of estimation and the more accurate assessment depends on practical observation of previous cases

**4-Turner syndrome has one X chromosome, choose the incorrect statement :**

a. Turner has a mild phenotype and patients are independent and can take care of themselves

b. Turner cases are mostly infertile

c. Turner syndrome is the only full monosomy that is seen in humans

d. What is  seen of Turner cases in only the tip of the iceberg, most cases aborted

e.  Monosomy X is of low effect, thus majority of conceptions continue to birth

**5- In 1954 we knew that human chromosomes were 46 as opposed to the previous belief that they were 48. What technical advance helped in that recognition?**

a. The use of chemical to fix cell division at metaphase

b. The use of new staining procedure

c. The use of mitogen to increase cell division

d. The use of hypotonic solution

e. The use of fluorescent microscope

**6- The most frequent autosomal reciprocal translocation in humans is between chromosomes :**

a. 9 and 22

b. 11 and 22

c. X and Y

d. 14 and 21

**7- Choose the correct statement :**

a. Most chromosomal abnormalities can be seen in older adults then live birth babies then blastocysts

b. Most chromosomal abnormalities can be seen in blastocysts then live birth babies then older adults

c. Chromosomal abnormalities , when checked in different intervals of the life of human , have the same frequency .

d. Most chromosomal abnormalities can be seen at birth stage, earlier than that are not as frequent

e. Aging individuals are expected to have the most of constitutional chromosomal abnormalities in their cells than other life stages

**8-In a translocation carrier, an exchange between chromosomes 3 and 6. One quarter of chromosome 3 went to chromosome 6 and one quarter of chromosome 6 went to chromosome 3. Choose the segregation mode most compatible with live birth :**

a. Adjacent 1

b. None

c. Interchange trisomy

d. Tertiary trisomy

e. Adjacent 2

**9- Choose the incorrect statement about imprinting :**

a. Imprinting is stable in cells, but can differ according to tissue type or developmental stage

b. Imprinting can be exposed by uniparental disomy

c. Imprinting needs to be re-programmed with each generation

d. None of the choices

e. Imprinting depends on the parent of origin of the chromosome

**10-Choose the correct statement :**

a. Sequential nondisjunction happen in both male and female gametogenesis simul

b. Complete nondisjunction will lead to trisomy

c. Simultaneous parental nondisjunction happens in meiosis I and II in same parent

d. None is correct

e. Multiple aneuploidies happen in theory but never seen in humans

**11-One of the pathogenic mechanisms that arise from chromosomal abnormalities is position effect, how?**

a. Can make a silent gene become active

b. Can convert an active gene into a silent one

c. Can cause decrease in gene expression

d. Can make a gene increase its expression

e. All choice are right

**12-A child patient was found to have an autosomal recessive disease. Mutation testing indicated homozygosity for the pathogenic mutation. Mutation testing in father found him heterozygous while the mother is wild type for both alleles. How could this happen?**

a. Child have inherited a chromosome from his paternal grandfather and the other homolog from his paternal grandmother

b. Child might be trisomic then trisomic rescue happened and resulted in paternal isodisomy

c. Child might be trisomic then trisomic rescue happened and resulted in maternal heterodisomy

d. Child might be trisomic then trisomic rescue happened and resulted in maternal Isodisomy

e. Child might be trisomic then trisomic rescue happened and resulted in paternal heterodisomy

**13-Human chromosomes are of types :**

a. All choices are right

b. Submetacentric , telocentric , acrocentric

c. Acrocentric , metacentric , submetacentric

d. Telocentric , acrocentric metacentric

e. Metacentric submetacentric , telocentric

**14- In Robertsonan , complete to happen resilt except :**

a. Translocations involving and 11

b. Translocations involving chromosomes 13 and 22

c. Translocations chromosomes 21 and 22

d. Translocations involving chromosomes 14 and 15

e. Translocations involving chromosomes 13 and 21

**15- Choose the most correct statement:**

a. (11;22) is the most common Robertsonian translocation

b. (14;21) is the most common Robertsonian translocation

c. (11;22) could lead to Down syndrome

d. (13;14) is the most common chromosomal rearrangement

e. (13;14) is the most common reciprocal translocation

**16-Choose the most correct statement :**

a. Heterosynapsis result in abnormal consequences in alignment of rearranged chromosomes

b. Heterosynapsis is alignment of homologous chromosomes regardless of their differences due to chromosomal rearrangement

c. All is correct

d. Heterosynapsis is alignment of nonhomologous chromosomes

e. Homosynapsis in rearranged chromosomes result in normal product of chromosome separation